

Current Work : Opinion and Limitations

- Numerical (Profile stats) + Textual Embeddings (BERT)
- Adversarial training, via GPT generated profiles (compared with manual fakes)
- Limitations- GPT profile is generated once (Generator not being updated), no usage of “network” of a profile, types of connections etc. It becomes more of a “plagiarism” task or resume skimming task - **ignoring the social network**
- Basically, assuming each interaction as an independent instance with side information encoded, into a supervised learning task
- Now, even actual people use LLMs to make Resumes/ LinkedIn profiles
LLM generated != Fake
- (similar paper) - playing around with Embeddings, with major limitation of datasets

[The Looming Threat of Fake and LLM-generated LinkedIn Profiles: Challenges and Opportunities for Detection and Prevention](#)

Better Parameters:

- Employment-path inconsistencies
(impossible job transitions / abnormal career graph)
- Weak or substitute social graphs
(sparse or unreciprocated cowork networks)
- Organization-level inconsistencies
(working for a company that doesn't exist / mismatched industry)
- Skill-to-role inconsistencies
(skills irrelevant to declared job history)
- Temporal inconsistencies
(job durations impossible or contradictory)
- Identity surface inconsistencies
(profile picture reuse/ AI pic)

Problem:

Lack of dataset on LinkedIn for “fake” profiles, and missing social graphs!!

Proposal:

1. Use large, benchmark graph datasets with human/bot labels:
 - [Twibot-22: Towards Graph-Based Twitter Bot Detection](#) (heterogeneous user–tweet–relation graph, 10k+ labeled users)
 - [GitHub - GraphDetec/MGTAB: A Multi-relational Graph-Based Twitter Account Detection Benchmark](#) (multi-modal bot detection benchmark with stance & sentiment features)
 - [Twitter Follower–Followee Sybil Graph](#) (graph with benign vs sybil labels)
2. Methodology
 - Designing a Knowledge Graph Schema that can be generalized across multiple datasets for Heterogeneous Graphs
 1. [Heterogeneous Graph Attention Network | The World Wide Web Conference](#)
 2. [KGAT: Knowledge Graph Attention Network for Recommendation Scalable Knowledge Graph Construction from Twitter](#) (**Twitter specific**)
 3. [Schema-adaptable Knowledge Graph Construction](#) (for evolving Schema)
 - Textual Data from Tweets/Posts merged with Social Networks via Embeddings in the same space (A LOT OF WORK done here)
 1. [Sentiment analysis of tweets using text and graph multi-views learning | Knowledge and Information Systems](#) (Textual + Network Embeddings)
 2. [Detecting bots in social-networks using node and structural embeddings](#) (Graph Embeddings from social networks)
 3. [DANES: Deep Neural Network Ensemble Architecture for Social and Textual Context-aware Fake News Detection](#) (content embeddings + network/social context)
 - Finally, adding explainability to GNNs based classification methods for human vs bot predictions
 1. [Explainable Graph Neural Networks via Structural Externalities | IJCAI](#) (decomposing into sub graphs)
 2. [\[2311.15112\] Everybody Needs a Little HELP: Explaining Graphs via Hierarchical Concepts](#) (again, subgraphs)

Novelty: Bot/Fake Profile detection that introduces explainability via Graph DL